

Missingness-Aware SOFA Reconstruction Across Two ICU Databases With a Locked Alert-Governance Evaluation in MIMIC-IV

Satya Venkata Ranga Janaki Sriram Mentey

Abstract—Electronic-health-record implementations of deterministic severity scores must distinguish an observed normal value from a value that was never observed, and alert evaluations must distinguish score generation from clinician interruption. We implemented availability-time Sequential Organ Failure Assessment reconstruction with explicit partial and insufficient-data states, versioned rules, evidence provenance, and a separate alert-governance layer. Component observability was audited in MIMIC-IV 3.1 and a seeded 8,000-stay eICU-CRD 2.0 sample. Governance was selected on development and calibration anchor-year groups and evaluated once on a locked MIMIC-IV test set (14,407 stays; 5,315 labeled positive). Respiratory and hepatic inputs were the least observable components in both sources. Governed sensitivity was 98.49% (95% confidence interval, 98.15–98.81%) versus 98.51% for threshold-only scoring, and interruptive emissions were 7.67% of the naive interruptive baseline (95% confidence interval, 7.54–7.79%). However, interruptive sensitivity was 71.36%, precision was 6.08%, and 57.4 interruptions were emitted per positive stay reached by an interruption. Post hoc analyses showed that requiring an alert at least 2 hours before onset reduced governed sensitivity to 54.43% and interruptive sensitivity to 10.89%. Against discharge sepsis codes, 97.69% of code-positive stays had a governed alert, but so did 80.46% of code-negative stays. These results characterize a retrospective research prototype, not clinical validity or utility. Explicit missingness, separate routing endpoints, and lead-time-gated reporting prevent a high loose-window detection rate from concealing data, timing, and interruption-burden limitations.

Index Terms—alert fatigue, clinical decision support, data completeness, eICU-CRD, electronic health records, MIMIC-IV, reproducibility, SOFA

I. INTRODUCTION

THE Sequential Organ Failure Assessment (SOFA) score combines respiratory, coagulation, hepatic, cardiovascular, neurologic, and renal measurements into a deterministic representation of organ dysfunction [1]. The scoring table is simple; reconstructing it from longitudinal electronic health record (EHR) data is not. Measurements arrive at different frequencies, documentation time may differ from physiological event time, drug doses use heterogeneous units, and the absence of a measurement can be mistaken for a normal result. EHR completeness is therefore a property of the measurement process and intended use, not merely the fraction of populated database fields [2], [3].

A second ambiguity appears after scoring. A threshold crossing, a retained alert record, and an interruptive notification

are different events. Treating them as interchangeable hides the mechanisms that create alert fatigue and prevents a meaningful accounting of clinician interruptions [4], [5], [6]. Evaluations of deployed sepsis decision support likewise show that detection performance alone does not establish workflow benefit [7], [8].

This paper evaluates a biomedical-informatics boundary rather than a new clinical score. The CURIE prototype reconstructs SOFA using only information available at each event time, reports missing components instead of silently assigning reassuring values, and applies a separately versioned governance policy to passive and interruptive outputs. The evaluation uses MIMIC-IV 3.1 [9], [10] and eICU-CRD 2.0 [11], [12]. Its contributions are:

- 1) an explicit data-state contract for deterministic EHR scoring, including availability time, evidence lineage, and abstention when required inputs are unobserved;
- 2) a cross-source audit showing which SOFA inputs the adapters can actually recover, with eICU-CRD used only to assess extraction portability;
- 3) a locked MIMIC-IV evaluation that separates threshold crossings, governed records, and interruptive emissions, with prespecified ablations;
- 4) post hoc timing and label-sensitivity analyses that quantify how much of the loose-window detection rate reflects actionable lead time and how specific stay-level alerting is against an independently coded label; and
- 5) a reproducibility contract linking the protocol, data/index version, label implementation, rule version, operating point, bootstrap seed, and every reported aggregate.

No language model generates, suppresses, or modifies an alert. We do not claim diagnostic accuracy, clinical validation, improved outcomes, production readiness, or regulatory readiness.

II. RELATED WORK

A. SOFA reconstruction and missing data

SOFA was designed for daily bedside charting [1], and serial SOFA carries prognostic information [13]. Retrospective studies commonly compute it from EHR extracts using shared concept code; the MIMIC Code Repository provides such SQL for MIMIC and was introduced to make derived concepts reproducible [14]. A recurring convention treats an unmeasured component as normal (a subscore of zero), which Lambden *et al.* identify as a source of bias in clinical trials and observational research [15]. Data-quality frameworks separate

conformance, completeness, and plausibility [3] and define completeness relative to a task [2]. Our scorer does not propose a new imputation method; it refuses to impute and propagates component-level missingness into every emitted signal so that downstream evaluation can condition on it.

B. Sepsis labels and construct overlap

Sepsis-3 operationalizes sepsis as suspected infection with an acute SOFA increase of at least two points [16], [17]. When the reference label is itself SOFA-derived, a SOFA-based detector shares part of its construct with the label, so high sensitivity partly reflects incorporation. Administrative definitions based on discharge codes, such as explicit sepsis codes or infection-plus-organ-dysfunction algorithms [18], [19], have different error profiles and diverge from clinical surveillance definitions [20]; in MIMIC, sepsis identification methods yield markedly different cohorts and outcomes [21]. We therefore use discharge codes only as a label-sensitivity analysis, never as an onset time or as validation.

C. Alert burden and governance mechanisms

Clinicians override most interruptive alerts [22], and repeated alerts increase fatigue [6], [4]. Human-factors guidance recommends reducing redundant and low-value interruptions [23], and health-system studies measure interruption burden directly [5]. Refractory periods, deduplication, and tiered delivery are established design patterns; we do not claim that any single mechanism is new. Our contribution is to version these mechanisms as a policy object separate from the score and to attribute burden to each mechanism by prespecified ablation on a locked test set.

D. Evaluation of sepsis decision support

Machine-learning early-warning systems have reported strong retrospective discrimination [24], [25], [26], and a public benchmark standardized evaluation on hourly ICU data [27]. External evaluation of a widely deployed proprietary model found substantially lower performance and a large alert burden [8], whereas a prospective multi-site study associated a deployed system with outcomes when alerts were acted upon [28]. Reporting guidance emphasizes intended use, data provenance, and transparent evaluation [29], [30]. The present study is positioned before such evaluations: it characterizes data availability, timing, and interruption burden of a deterministic pipeline so that a later prospective study can be designed around measurable quantities.

III. SYSTEM AND METHODS

A. System boundary and information flow

Fig. 1 and Table I define the evaluated stages. The separation is intentional: a scorer answers what can be computed from the observed record; governance answers how a completed signal is routed. An optional language-model component in the larger prototype was disabled and is outside the alert path.

Poison or unsupported input is rejected to a dead-letter queue rather than converted to a normal clinical value. Python

TABLE I
DETERMINISTIC PROCESSING BOUNDARY.

Stage	Input/output	Invariant
Adapter	Source rows → typed events	Units and source times are retained; unsupported values remain unknown.
Timeline	Typed events → state	Only observations available by the scoring time are visible.
Scorer	State → SOFA signal	Missing inputs yield component-level missingness and partial status, not an assumed score of zero.
Governance	Signal → watch/page decision	Routing may delay or deduplicate a complete signal but cannot change its score or evidence.
Episode layer	Decisions episodes →	Repeated signals are grouped without erasing the underlying records.

TABLE II
MIMIC-IV COHORT FLOW (PROTOCOL V2 SELECTION).

Step	ICU stays
ICU stays in source	94,458
Excluded: invalid admission/discharge time	14
Excluded: ICU duration < 4 h	714
Excluded: later eligible stay, same admission	8,689
Protocol cohort	85,041
Not assigned: anchor-year group 2020–2022	9,566
Study cohort	75,475
Development (2008–2013)	44,755
Calibration (2014–2016)	16,313
Locked test (2017–2019)	14,407

reference scorers and Java/Flink streaming implementations are checked against shared golden fixtures, and a rule bundle cannot be activated unless this parity gate passes.

B. Datasets, cohort, and split policy

MIMIC-IV 3.1 contains deidentified data from a single US academic medical center; eICU-CRD 2.0 contains deidentified data from more than 200 US hospitals. Both require credentialed PhysioNet access and a signed data-use agreement. The frozen protocol included adults, required valid ICU admission and discharge times and an ICU duration of at least four hours, and retained the earliest eligible ICU stay per hospital admission. MIMIC's deidentification shifts calendar dates independently per patient; development, calibration, and test assignment therefore used each patient's `anchor_year_group` (2008–2010 and 2011–2013, 2014–2016, and 2017–2019, respectively) rather than shifted calendar years. Because the group is a patient attribute, the splits are patient-disjoint; because it locates only the patient's anchor year, the temporal ordering of individual stays is approximate. Table II gives every denominator.

An earlier cohort audit selected the first ICU stay per admission before the time filters and retained 84,855 stays. All 84,855 are in the protocol cohort; the protocol rule adds 186 admissions whose first ICU stay was invalid or shorter than four hours but whose later stay was eligible. The cohort-wide completeness audit below uses the earlier 84,855-stay selection. eICU-CRD was sampled with a fixed seed (8,000 stays, seed 42) for an extraction-completeness analysis only.

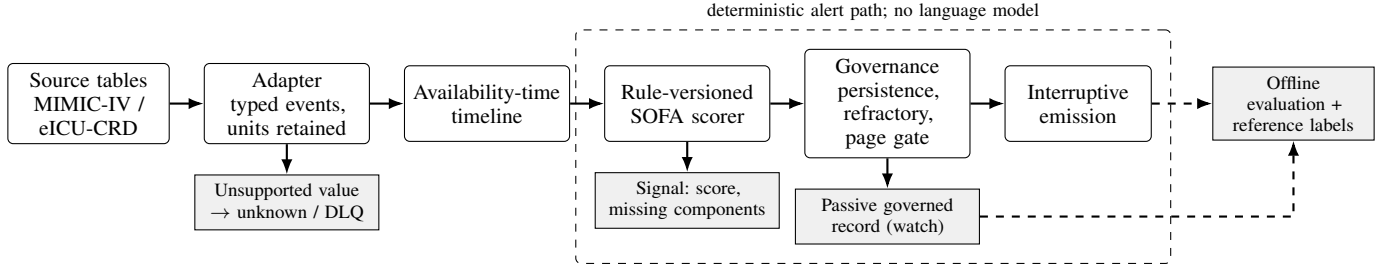


Fig. 1. Information flow. Each emitted signal carries its score, component values, missing-component list, rule version, and source-observation identifiers. Governance may delay, deduplicate, or demote a signal to a passive record but cannot change its score or evidence. Reference labels are joined to the outputs only offline, for evaluation (dashed arrows), and are never visible to the timeline or scorer.

C. Availability-time SOFA reconstruction

At each incoming event time, the timeline selects the most recent usable value within the component’s documented lookback rule. A value is usable only after the time at which it would have been available to the scoring system; laboratory and charted values defer to their storage time, which prevents a later result or correction from entering an earlier score. When no usable value exists, the component is marked missing, and the total is emitted with its completeness state; it is never represented as a fully observed score.

Respiration prioritizes $\text{PaO}_2/\text{FiO}_2$. A $\text{SpO}_2/\text{FiO}_2$ fallback uses the published relationship only within its supported saturation range [31]; FiO_2 is never inferred from missing oxygen documentation. Cardiovascular processing preserves the source vasopressor unit and converts only combinations covered by an explicit unit policy. Vasopressin rates that cannot be converted without an unavailable concentration remain unknown. Urine-output windows require coverage for the specified duration. These choices make missingness visible but do not establish that a reconstructed component is clinically correct.

D. Completeness and portability audit

For each SOFA input we measured, cohort-wide in MIMIC-IV, the fraction of stays with at least one usable observation and the fraction observed during the first ICU day. In seeded 8,000-stay samples from each database we replayed the scorer and recorded, per component, the fraction of stays whose final score of the stay lacked a usable value within that component’s lookback window. We also reconciled indexed-loader output against a direct source-table scan for a deterministic 25-stay sample. The eICU audit exercised the same typed-event and scoring contracts through source-specific extraction mappings. It did not evaluate labels, governance, outcomes, or cross-site clinical performance.

E. Governance policy and locked evaluation

The governance layer maintains trajectory persistence, refractory state, and episode membership. A governed record is a retained, non-duplicative alert event. An interruptive emission additionally satisfies the page gate, which can require repeated crossings, sustained elevation, and a rising trajectory. The selected policy used a 90-min refractory interval and a page

gate requiring two crossings sustained for 30 min with a rising score. Removing or delaying an interruption does not alter the underlying signal.

The protocol, candidate policies, and ablations were specified and frozen before locked test evaluation. Candidates were developed on development groups, one operating point was selected on calibration (minimum interruptive ratio subject to the sensitivity criterion), and the test split was evaluated once. The primary detection window was 12 h before through 6 h after the Sepsis-3-aligned onset generated from a pinned MIMIC Code Repository revision [14], [17]. The primary endpoints were governed sensitivity (success if within 10 percentage points of threshold-only sensitivity or at least 70%) and the ratio of interruptive emissions under governance to the naive interruptive baseline (success if at most 25%). This endpoint is not the ratio of all governed records to all raw threshold crossings; both are reported.

Sensitivity is the fraction of labeled-positive stays with at least one qualifying event in the window. Burden is reported as counts and rates per 100 patient-days. Interruptive precision is the fraction of interruptive emissions that fall inside a labeled-positive stay’s detection window. Number needed to alert (NNA) is interruptive emissions divided by labeled-positive stays with at least one in-window interruptive emission (3,793 under full governance); it is a burden statistic, not a treatment effect.

F. Post hoc secondary analyses

After the locked evaluation, and without changing the operating point, we replayed the test split once more with the frozen policy and defined three secondary analyses. None was used for selection.

Lead-time gate. A labeled-positive stay counts as detected only if an alert occurs in $[t_{\text{onset}} - 12 \text{ h}, t_{\text{onset}} - 2 \text{ h}]$. This removes credit for alerts coincident with, or after, the label.

Discharge-code label sensitivity. A stay is code-positive if its hospital admission carries at least one code from a frozen ICD-9/10 list of explicit sepsis, severe-sepsis, and septic-shock codes (ICD-9 038.x, 995.91, 995.92, 785.52; ICD-10 A40–A41, R65.20, R65.21, O85). Because discharge codes have no onset time, we report the fraction of code-positive stays with any alert during the ICU stay, alongside the same fraction in code-negative stays. Codes were never features and were joined only after replay.

TABLE III

COHORT CHARACTERISTICS BY SPLIT. VALUES ARE N, N (%), OR MEDIAN [INTERQUARTILE RANGE].

	Development	Calibration	Test
ICU stays	44,755	16,313	14,407
Patients	31,223	13,157	12,165
Age, y	64 [52–75]	65 [53–76]	66 [54–76]
Female	20,305 (45.4)	7,123 (43.7)	6,228 (43.2)
ICU stay, d	1.85 [1.05–3.43]	1.92 [1.11–3.69]	2.03 [1.14–4.01]
Sepsis-3 positive	21,484 (48.0)	7,426 (45.5)	5,315 (36.9)

Subgroups. Detection and burden were summarized by sex and age band. These summaries are descriptive and not powered for between-group inference.

G. Statistics and reproducibility

Confidence intervals are stay-level percentile bootstrap intervals (1,000 replicates, seed 42). Every number in this paper resolves to one of two content-hashed sidecars: the locked study manifest (primary endpoints, ablations, bootstrap intervals, miss count) and a publication-aggregate sidecar (cohort flow, characteristics, completeness, post hoc analyses, and subgroups). The latter records the SHA-256 of each audit input, the index hash, the label-artifact hash, and the operating-point hash, and reproduces the frozen governed sensitivity from the replay rows before it can be written. Cells describing fewer than 11 stays are suppressed.

IV. RESULTS

A. Cohort

Table III describes the three splits. Median age and sex distribution were similar across splits, whereas Sepsis-3 label prevalence was lower in the locked test groups (36.9%) than in development (48.0%) and calibration (45.5%). The test split contained 14,407 stays, 5,315 labeled-positive stays, and 54,401 patient-days.

B. Component observability

Table IV reports input observability and scored component missingness. Cardiovascular, neurologic, coagulation, and renal components were missing in fewer than 5% of scored stays in both databases. The respiratory component was missing in 66.4% of scored MIMIC-IV stays and 59.7% of scored eICU-CRD stays, driven by sparse FiO₂ documentation, and the hepatic component in 13.4% and 27.6%, respectively. The final score of the stay was fully observed in 30.2% (MIMIC-IV) and 29.8% (eICU-CRD) of sampled stays. Observability describes the available record; it does not show that observed components were accurate.

In the MIMIC-IV vasopressor audit, 685,045 infusion rows mapped to a supported dose representation and 31,020 remained explicitly unknown, of which 30,559 were vasopressin rates recorded in units per hour without a usable concentration. The source-to-index reconciliation sample had no missing or extra rows. These engineering checks reduce silent transformation error; they are not a comparison with bedside adjudication.

TABLE IV

INPUT OBSERVABILITY (%), MIMIC-IV COHORT-WIDE, 84,855 STAYS) AND COMPONENT MISSINGNESS IN THE FINAL SCORE OF EACH STAY (%), SEEDED 8,000-STAY SAMPLES).

Component	Input	MIMIC input observed		Component missing	
		Any	Day 1	MIMIC	eICU
Respiration	PaO ₂	70.6	53.8	66.4	59.7
	FiO ₂	49.3	45.7		
Coagulation	Platelets	99.7	96.5	0.1	4.1
	Bilirubin	80.9	44.0	13.4	27.6
Cardiovascular	MAP	99.8	99.5	0.2	1.7
CNS	GCS	99.8	99.7	0.2	0.9
Renal	Creatinine	99.8	97.2	<0.1	1.8
	Urine output	96.5	95.5		
Fully observed final score				30.2	29.8

TABLE V

LOCKED MIMIC-IV TEST RESULTS (14,407 STAYS). CI DENOTES THE PRESPECIFIED STAY-LEVEL BOOTSTRAP INTERVAL; INTERVALS WERE PRESPECIFIED ONLY FOR THE ROWS THAT SHOW ONE.

Metric	Result
Threshold-only sensitivity	98.51%
Governed sensitivity (95% CI)	98.49% (98.15–98.81)
Interruptive sensitivity	71.36%
Raw threshold crossings	4,351,252
Governed records	784,548
Governed/raw record ratio	18.03%
Naive interruptive emissions	2,840,825
Governed interruptive emissions	217,778
Interruptive ratio (95% CI)	7.67% (7.54–7.79)
Governed records per 100 patient-days	1,442.2
Interruptions per 100 patient-days	400.3
Interruptive precision	6.08%
Interruptive NNA (95% CI)	57.4 (55.6–59.4)

C. Locked MIMIC-IV governance evaluation

Table V separates the output levels. Both primary endpoints were met. The all-record reduction was 784,548 governed records divided by 4,351,252 threshold crossings, or 18.03%. The prespecified interruptive endpoint was 217,778 divided by 2,840,825 naive interruptive emissions, or 7.67% (95% CI, 7.54–7.79%).

Governance retained 5,235 of 5,315 labeled-positive stays within the loose window, leaving 80 without a governed detection (1.51%). The near-equivalence of governed and threshold-only sensitivity applies to passive governed records. It does not apply to clinician interruption: the page gate reduced interruptive sensitivity to 71.36% while still producing 400.3 interruptions per 100 patient-days.

D. Prespecified ablations

Table VI attributes burden to mechanisms. The refractory interval was the only deduplicating mechanism: removing it returned governed records to the raw crossing count and raised interruptive emissions from 217,778 to 1,555,348. Removing the page gate more than doubled interruptive emissions (469,718) and raised interruptive sensitivity to 78.53% at lower precision (4.45%). Six ablations produced results identical to full governance because the mechanism was already inactive in the selected policy (trajectory persistence, multiple crossings,

TABLE VI

PRESPECIFIED ABLATIONS ON THE LOCKED TEST SPLIT. SENS., SENSITIVITY (%); INT., INTERRUPTIVE; RATIO, INTERRUPTIVE EMISSIONS RELATIVE TO THE NAIVE INTERRUPTIVE BASELINE (%).

Policy	Gov. sens.	Int. sens.	Records	Int. Ratio	NNA
Threshold-only	98.51	80.30	4,351,252	2,840,825	100.0
Full governance	98.49	71.36	784,548	217,778	7.67
—Refractory	98.51	74.98	4,351,252	1,555,348	54.75
—Page gate	98.49	78.53	784,548	469,718	16.53
Six others ^a	98.49	71.36	784,548	217,778	7.67

^aPersistence, crossings, baseline, context suppression, episode arbitration, late-event buffer.

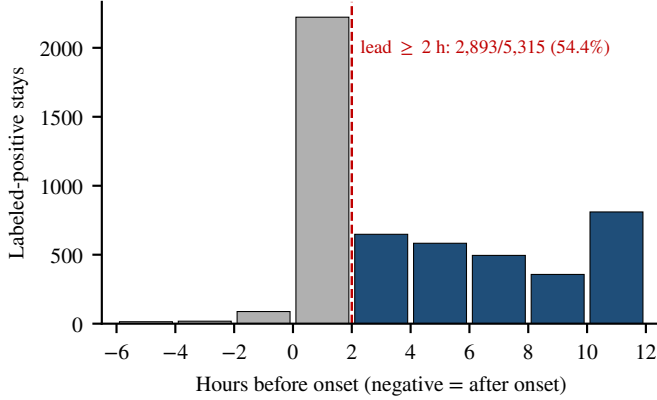


Fig. 2. Lead of the earliest governed alert relative to Sepsis-3-aligned onset among 5,235 detected test stays. Grey bars fall short of the 2-h lead gate. The rightmost bin is right-censored by the 12-h detection window.

baseline delta) or did not alter any emission in this replay (context suppression, episode arbitration, late-event buffer).

E. Lead time and label sensitivity (post hoc)

Fig. 2 shows the lead of the earliest governed alert in the detection window. The median lead among the 5,235 detected stays was 2.68 h (interquartile range, 0.85–7.23 h), and in 120 stays the earliest in-window alert followed onset. Requiring at least 2 h of lead reduced governed sensitivity from 98.49% to 54.43% and interruptive sensitivity from 71.36% to 10.89% (Table VII). Thus, nearly half of the loose-window detections occurred within 2 h of, or after, the label.

Discharge codes and Sepsis-3 labels overlapped only partially: 1,843 stays were positive by both, 406 by codes only, and 3,472 by Sepsis-3 only. Nearly all code-positive stays received a governed alert, but so did 80.46% of code-negative stays. Interruptive alerts separated the groups more (82.13% versus 51.77%) yet still reached half of code-negative stays. Stay-level alerting is therefore sensitive but weakly specific against an independently assigned label.

F. Subgroups

Table VIII summarizes detection and burden by sex and age. Loose-window governed sensitivity was at least 97.8% in every subgroup. Lead-gated sensitivity was lower in women than in men (51.67% versus 56.39%) and in patients aged 65 years or older than in younger patients. Interruptive sensitivity

TABLE VII

POST HOC TIMING AND LABEL-SENSITIVITY ANALYSES ON THE LOCKED TEST SPLIT. VALUES ARE % (95% STAY-BOOTSTRAP CI) WHERE SHOWN; ROWS WITHOUT AN INTERVAL ARE DESCRIPTIVE POINT ESTIMATES THAT WERE NOT BOOTSTRAPPED.

Analysis	Result
<i>Sepsis-3 onset (5,315 positive stays)</i>	
Governed, primary window	98.49 (98.15–98.81)
Governed, lead ≥ 2 h	54.43 (53.02–55.79)
Interruptive, primary window	71.36
Interruptive, lead ≥ 2 h	10.89 (10.08–11.76)
<i>Discharge sepsis codes (2,249 positive stays)</i>	
Any governed alert, code-positive	97.69 (97.05–98.26)
Any governed alert, code-negative (12,158)	80.46
Any interruptive alert, code-positive	82.13 (80.46–83.60)
Any interruptive alert, code-negative	51.77
Code- and Sepsis-3-positive (1,843): primary window	99.62
Code- and Sepsis-3-positive: lead ≥ 2 h	58.82
<i>Sepsis-3-negative stays (9,092)</i>	
Any governed alert	73.58
Any interruptive alert	40.54

TABLE VIII

DESCRIPTIVE SUBGROUP RESULTS ON THE LOCKED TEST SPLIT. SENS., SENSITIVITY (%); LEAD, GOVERNED SENSITIVITY WITH LEAD ≥ 2 h (%); INT./100 PD, INTERRUPTIONS PER 100 PATIENT-DAYS.

Group	Stays	Pos.	Gov. sens.	Lead	Int. sens.	Int./100 pd
Female	6,228	2,210	97.87	51.67	67.29	372.5
Male	8,179	3,105	98.94	56.39	74.27	420.4
18–44 y	2,038	641	97.97	60.22	71.61	346.0
45–64 y	4,772	1,718	98.72	58.15	74.80	392.0
65–79 y	5,113	1,877	98.77	50.83	72.14	423.6
≥ 80 y	2,484	1,079	97.96	51.34	64.41	413.6

was lowest in patients aged 80 years or older (64.41%). These differences are descriptive; they may reflect differences in measurement frequency or label timing and warrant targeted analysis before any deployment.

V. DISCUSSION

This study supports four technical conclusions. First, a deterministic score does not eliminate data-quality uncertainty. Explicit component states make the uncertainty inspectable: fewer than one third of sampled stays ended with a fully observed SOFA score in either database, almost entirely because of respiratory and hepatic inputs. A pipeline that silently scored those components as normal would report complete scores for every stay and bias severity downward exactly where documentation is sparse [15].

Second, score generation and clinician interruption require separate endpoints. Governance preserved loose-window detection while emitting 7.67% as many interruptions, and the ablations localize this effect to two mechanisms: refractory deduplication and the page gate. Yet interruptive precision remained 6.08% and 400.3 interruptions were emitted per 100 patient-days. Reporting only the sensitivity or only the reduction ratio would obscure one of these facts.

Third, loose detection windows overstate actionable warning. The primary window credits alerts up to 6 h after onset. When

at least 2 h of lead was required, governed sensitivity fell to 54.43% and interruptive sensitivity to 10.89%. Part of this gap is expected from construct overlap: the Sepsis-3-aligned onset is defined by the same SOFA rise that triggers the score, so many alerts coincide with the label by construction. The discharge-code analysis addresses a different question and gives a complementary warning: alerts reached nearly every code-positive stay but also most code-negative stays. Neither analysis validates the detector; both bound what the primary sensitivity can be taken to mean. We recommend that governance evaluations report lead-gated and label-negative results next to any loose-window sensitivity.

Fourth, reproducibility for a rule-based informatics pipeline is more than publishing source code. The results depend on source version, extraction semantics, availability time, label implementation, cohort rule, split, rule bundle, operating point, and metric denominator. A one-rule difference in stay selection changed the cohort by 186 stays; recording both rules and reconciling them exactly is what makes that difference auditable. The evidence package therefore freezes these objects with content hashes, and corrections are issued as new versions rather than edits.

A. Relation to the companion manuscript

A separate manuscript, “Governing Interruptions After Deterministic Deterioration Scoring: A Retrospective Evaluation on PhysioNet Challenge 2019,” is under review at the *International Journal of Medical Informatics* (manuscript IJMEDI-S-26-06829). It evaluates governance of a partial SOFA score on the hourly PhysioNet Challenge 2019 benchmark [27]. The present work does not reuse that cohort, its holdout results, tables, or figures. Its primary contributions are the EHR reconstruction and provenance boundary, the cross-database completeness audit, and a locked MIMIC-IV evaluation with timing and label-sensitivity analyses.

VI. LIMITATIONS

The study is retrospective and offline. The primary onset label is SOFA-derived, creating construct overlap with the score under evaluation; the reported sensitivity is therefore a pipeline-behavior measure, not diagnostic accuracy. The lead-time, discharge-code, and subgroup analyses were defined after the locked evaluation and should be read as post hoc. Discharge codes reflect billing and documentation practice, carry no onset time, and are themselves an imperfect reference [20], [21]. No clinician adjudication, prospective shadow deployment, human-factors assessment, treatment endpoint, mortality endpoint, or comparison with a deployed commercial system was performed. eICU-CRD was not used for labeled outcome or governance evaluation, so this paper offers no external validation of detection or alerting behavior. The cohort-wide completeness audit used an earlier stay-selection rule that differs from the study cohort by 186 of 85,041 stays. Component observability does not establish measurement validity, and differing documentation practices limit direct cross-database comparison. MIMIC-IV reflects a single center, and label prevalence differed across anchor-year groups.

VII. CONCLUSION

An EHR scoring pipeline should expose what it did not observe, and an alerting evaluation should distinguish what it computed, retained, and sent interruptively, and when. In a locked MIMIC-IV evaluation, the CURIE research prototype preserved loose-window governed detection while reducing the prespecified interruptive burden ratio to 7.67%, but interruptive precision, absolute notification burden, and lead-gated sensitivity remained limiting. A two-database completeness audit localized missing-input risk without treating availability as clinical validity. These findings motivate prospective workflow and clinical evaluation; they do not provide it.

AUTHOR CONTRIBUTIONS

S. V. R. J. S. Mentey: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, and writing.

DATA AND CODE AVAILABILITY

MIMIC-IV 3.1 and eICU-CRD 2.0 are available to credentialed users through PhysioNet under their data-use agreements; patient-level data and replay outputs are not redistributed. Source code, rule bundles, parity fixtures, tests, the frozen protocol, operating point, study manifest, and publication-aggregate sidecar with content hashes, and reproduction commands are available at <https://github.com/sreeram843/curie-prediction-pipeline>.

ETHICS STATEMENT

This study is a secondary analysis of deidentified data obtained from PhysioNet under credentialed-access data-use agreements, with no new data collection and no contact with participants. Because the analysis uses only deidentified data and involves no interaction with participants, it did not require independent ethics review. No new consent was sought; the source databases were created under approvals that waived individual consent: MIMIC-IV was approved by the institutional review boards of Beth Israel Deaconess Medical Center and the Massachusetts Institute of Technology with a waiver of consent, and eICU-CRD was exempt from institutional review because of its retrospective design and certified deidentification.

FUNDING

This research received no external funding and was conducted on a self-funded basis.

COMPETING INTERESTS

The author declares no competing interests.

ACKNOWLEDGMENT

Generative AI tools were used in this work under the author's direction: OpenAI Codex, the Cursor coding agent (using large language models from multiple vendors), and Anthropic Claude Code. They assisted with software implementation, including data adapters, evaluation and aggregation code, and tests; with figure-generation scripts; and with manuscript organization, language drafting, and LaTeX editing across all sections. The author designed the study, specified the protocol, reviewed all generated code and text, verified every numerical statement against the frozen repository artifacts, and accepts full responsibility for the manuscript.

REFERENCES

- [1] J.-L. Vincent *et al.*, "The SOFA (Sepsis-related Organ Failure Assessment) score to describe organ dysfunction/failure," *Intensive Care Med.*, vol. 22, pp. 707–710, 1996.
- [2] N. G. Weiskopf and C. Weng, "Methods and dimensions of electronic health record data quality assessment: Enabling reuse for clinical research," *J. Am. Med. Inform. Assoc.*, vol. 20, no. 1, pp. 144–151, 2013.
- [3] M. G. Kahn *et al.*, "A harmonized data quality assessment terminology and framework for the secondary use of electronic health record data," *eGEMS*, vol. 4, no. 1, art. no. 1244, 2016.
- [4] P. J. Embi and A. C. Leonard, "Evaluating alert fatigue over time to EHR-based clinical trial alerts: Findings from a randomized controlled study," *J. Am. Med. Inform. Assoc.*, vol. 19, pp. e145–e148, 2012.
- [5] P. Elias *et al.*, "Evaluating the impact of interruptive alerts within a health system," *Appl. Clin. Inform.*, vol. 10, pp. 909–917, 2019.
- [6] J. S. Ancker *et al.*, "Effects of workload, work complexity, and repeated alerts on alert fatigue in a clinical decision support system," *BMC Med. Inform. Decis. Mak.*, vol. 17, art. no. 36, 2017.
- [7] N. L. Downing *et al.*, "Electronic health record-based clinical decision support alert for severe sepsis: A randomised evaluation," *BMJ Qual. Saf.*, vol. 28, pp. 762–768, 2019.
- [8] A. Wong *et al.*, "External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients," *JAMA Intern. Med.*, vol. 181, no. 8, pp. 1065–1070, 2021.
- [9] A. Johnson *et al.*, "MIMIC-IV," version 3.1, PhysioNet, 2024, doi: 10.13026/kpb9-mt58.
- [10] A. E. W. Johnson *et al.*, "MIMIC-IV, a freely accessible electronic health record dataset," *Sci. Data*, vol. 10, art. no. 1, 2023, doi: 10.1038/s41597-022-01899-x.
- [11] T. Pollard, A. Johnson, J. Raffa, L. A. Celi, O. Badawi, and R. Mark, "eICU Collaborative Research Database," version 2.0, PhysioNet, 2019, doi: 10.13026/C2WM1R.
- [12] T. J. Pollard *et al.*, "The eICU Collaborative Research Database, a freely available multi-center database for critical care research," *Sci. Data*, vol. 5, art. no. 180178, 2018, doi: 10.1038/sdata.2018.178.
- [13] F. L. Ferreira, D. P. Bota, A. Bross, C. Mélot, and J.-L. Vincent, "Serial evaluation of the SOFA score to predict outcome in critically ill patients," *JAMA*, vol. 286, no. 14, pp. 1754–1758, 2001.
- [14] A. E. W. Johnson, D. J. Stone, L. A. Celi, and T. J. Pollard, "The MIMIC Code Repository: Enabling reproducibility in critical care research," *J. Am. Med. Inform. Assoc.*, vol. 25, no. 1, pp. 32–39, 2018.
- [15] S. Lambden, P. F. Laterre, M. M. Levy, and B. Francois, "The SOFA score—Development, utility and challenges of accurate assessment in clinical trials," *Crit. Care*, vol. 23, art. no. 374, 2019.
- [16] M. Singer *et al.*, "The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3)," *JAMA*, vol. 315, no. 8, pp. 801–810, 2016.
- [17] C. W. Seymour *et al.*, "Assessment of clinical criteria for sepsis: For the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3)," *JAMA*, vol. 315, no. 8, pp. 762–774, 2016.
- [18] D. C. Angus *et al.*, "Epidemiology of severe sepsis in the United States: Analysis of incidence, outcome, and associated costs of care," *Crit. Care Med.*, vol. 29, no. 7, pp. 1303–1310, 2001.
- [19] G. S. Martin, D. M. Mannino, S. Eaton, and M. Moss, "The epidemiology of sepsis in the United States from 1979 through 2000," *N. Engl. J. Med.*, vol. 348, no. 16, pp. 1546–1554, 2003.
- [20] C. Rhee *et al.*, "Incidence and trends of sepsis in US hospitals using clinical vs claims data, 2009–2014," *JAMA*, vol. 318, no. 13, pp. 1241–1249, 2017.
- [21] A. E. W. Johnson *et al.*, "A comparative analysis of sepsis identification methods in an electronic database," *Crit. Care Med.*, vol. 46, no. 4, pp. 494–499, 2018.
- [22] H. van der Sijs, J. Aarts, A. Vulto, and M. Berg, "Overriding of drug safety alerts in computerized physician order entry," *J. Am. Med. Inform. Assoc.*, vol. 13, no. 2, pp. 138–147, 2006.
- [23] S. Phansalkar *et al.*, "A review of human factors principles for the design and implementation of medication safety alerts in clinical information systems," *J. Am. Med. Inform. Assoc.*, vol. 17, no. 5, pp. 493–501, 2010.
- [24] K. E. Henry, D. N. Hager, P. J. Pronovost, and S. Saria, "A targeted real-time early warning score (TREWScore) for septic shock," *Sci. Transl. Med.*, vol. 7, no. 299, art. no. 299ra122, 2015.
- [25] S. Nemati *et al.*, "An interpretable machine learning model for accurate prediction of sepsis in the ICU," *Crit. Care Med.*, vol. 46, no. 4, pp. 547–553, 2018.
- [26] M. Moor, B. Rieck, M. Horn, C. R. Jutzeler, and K. Borgwardt, "Early prediction of sepsis in the ICU using machine learning: A systematic review," *Front. Med.*, vol. 8, art. no. 607952, 2021.
- [27] M. A. Reyna *et al.*, "Early prediction of sepsis from clinical data: The PhysioNet/Computing in Cardiology Challenge 2019," *Crit. Care Med.*, vol. 48, no. 2, pp. 210–217, 2020.
- [28] R. Adams *et al.*, "Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis," *Nat. Med.*, vol. 28, pp. 1455–1460, 2022.
- [29] G. S. Collins *et al.*, "TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, art. no. e078378, 2024.
- [30] J. Wiens *et al.*, "Do no harm: A roadmap for responsible machine learning for health care," *Nat. Med.*, vol. 25, pp. 1337–1340, 2019.
- [31] T. W. Rice *et al.*, "Comparison of the SpO₂/FiO₂ ratio and the PaO₂/FiO₂ ratio in patients with acute lung injury or ARDS," *Chest*, vol. 132, no. 2, pp. 410–417, 2007.